Challenges Faced

- **Dataset Preparation:** Creating and maintaining a structured hashtag dataset with date, platform, region, and hashtag information was required for complete analysis.

- **Spark Configuration:** Setting up Apache Spark and PySpark correctly and ensuring the Spark session worked without configuration errors was a challenge.

- **Hashtag Grouping and Counting:** Grouping identical hashtags and calculating their frequencies required proper use of Spark DataFrame operations.

- **Trend Analysis:** Comparing hashtag popularity across multiple days required organizing the data date-wise and handling missing hashtag occurrences.

- **Regional and Category Analysis:** Analyzing hashtags by region and classifying them into Technology, Sports, and Entertainment categories required additional transformations.

- **Visualization:** Generating and saving the three graphs using Matplotlib required converting Spark DataFrames into a format suitable for plotting.

- **Large-Scale Data Handling:** Understanding how Spark processes and analyzes large datasets efficiently was an important practical challenge.